TEST REPORT

Applicant:	Pinout LTD
Address of Applicant:	Evripidou 12A, Agia Zoni, 3031, Limassol, Cyprus
Manufacturer/Factory:	Pinout LTD
Address of Manufacturer/Factory:	Evripidou 12A, Agia Zoni, 3031, Limassol, Cyprus

Equipment Under Test (EUT)

Product Name:	Altruist Urban
Model No.:	ES-ALTRUIST-URB-1
Trade Mark:	Hardware for Cyberpunks
Applicable standards:	EN 55032:2015+A1:2020 EN 55035:2017+A11:2020 EN IEC 61000-3-2:2019+A1:2021 EN 61000-3-3:2013+A2:2021
Date of sample receipt:	January 22, 2026
Date of Test:	January 22- February 11, 2026
Date of report issued:	February 11, 2026
Test Result :	PASS *

* In the configuration tested, the EUT complied with the standards specified above.

Authorized Signature:

Robinson Luo
Laboratory Manager

This results shown in this test report refer only to the sample(s) tested, this test report cannot be reproduced, except in full, without prior written permission of the company. The report would be invalid without specific stamp of test institute and the signatures of compiler and approver.

2 Version

Version No.	Date	Description
00	*February 11, 2026*	*Original*

Test by: ***Date:*** *February 11, 2026*

Test Engineer

Prepared by: ***Date:*** *February 11, 2026*

Project Engineer

Reviewed by: ***Date:*** *February 11, 2026*

Reviewer

3 Contents

1 COVER PAGE 1

2 VERSION 2

3 CONTENTS 3

4 TEST SUMMARY 4

5 GENERAL INFORMATION 5

5.1 GENERAL DESCRIPTION OF EUT 5
5.2 TEST MODE AND TEST VOLTAGE 5
5.3 DESCRIPTION OF SUPPORT UNITS 5
5.4 MONITORING OF EUT FOR ALL IMMUNITY TEST 5
5.5 DEVIATION FROM STANDARDS 5
5.6 ABNORMALITIES FROM STANDARD CONDITIONS 5
5.7 TEST FACILITY 6
5.8 TEST LOCATION 6

6 TEST INSTRUMENTS LIST 7

7 EMISSION TEST RESULTS 10

7.1 RADIATED EMISSION 10
7.2 CONDUCTED EMISSION 16
7.3 HARMONIC EMISSION 19
7.4 FLICKER EMISSION 19

8 IMMUNITY TEST RESULTS 20

8.1 PERFORMANCE CRITERIA DESCRIPTION IN CLAUSE 8 OF EN 55035 20
8.2 ELECTROSTATIC DISCHARGE 21
8.3 CONTINUOUS RF ELECTROMAGNETIC FIELD DISTURBANCES 23
8.4 ELECTRICAL FAST TRANSIENTS 25
8.5 SURGES 27
8.6 CONTINUOUS INDUCED RF DISTURBANCE 29
8.7 VOLTAGE DIPS AND VOLTAGE INTERRUPTIONS 30

9 TEST SETUP PHOTO 31

10 EUT CONSTRUCTIONAL DETAILS 35

4 Test Summary

Test item	Test Requirement	Test Method	Class / Severity	Result
Radiated Emission (up to 1G)	EN 55032	EN 55032	Class B	Pass
Radiated Emission (Above 1G) #	EN 55032	EN 55032	Class B	Pass
Conducted Emission	EN 55032	EN 55032	Class B	Pass
Asymmetric mode conducted emissions	EN 55032	EN 55032	Class B	N/A
Disturbance voltage at antenna terminal	EN 55032	EN 55032	Class B	N/A
Harmonic Emission	EN IEC 61000-3-2	EN IEC 61000-3-2	Class A	N/A
Flicker Emission	EN 61000-3-3	EN 61000-3-3	Clause 5	Pass
Electrostatic discharge	EN 55035	EN 61000-4-2	Contact ±4 kV Air ±2,±4,±8 kV	Pass
Continuous RF electromagnetic field disturbances	EN 55035	EN 61000-4-3	3V/m 80%, 1kHz, AM	Pass
Electrical fast transients	EN 55035	EN 61000-4-4	± 1.0kV for AC port; ± 0.5kV for signal ports	Pass
Surges	EN 55035	EN 61000-4-5	AC port: 1kV Line to Line, 2kV line to earth; Signal port: 1kV	Pass
Continuous induced RF disturbance	EN 55035	EN 61000-4-6	0.15MHz to 10MHz:3V 10MHz to 30MHz:3V to 1V 30MHz to 80MHz:1V 80%, 1kHz Amp. Mod.	Pass
Voltage dips and Voltage interruptions	EN 55035	EN 61000-4-11	0 % UT* for 0.5per 0 % UT* for 250per 70 % UT* for 25per	Pass

Remark:

1. *Pass: Comply with the essential requirements in the standard.*
2. *N/A: Not applicable*
3. *U_T: the nominal supply voltage; D.M: Differential Mode; C.M: Common Mode.*
4. *# Refer to EN 55032 clause 8 conditional testing procedure :*

Highest internal frequency (Fx)	*Highest measured frequency*
Fx ≤ 108 MHz	*1GHz*
108 MHz < Fx ≤ 500 MHz	*2GHz*
500 MHz < Fx ≤ 1 GHz	*5GHz*
Fx > 1 GHz	*5 x Fx up to a maximum of 6 GHz*
NOTE 1 For FM and TV broadcast receivers, Fx is determined from the highest frequency generated or used excluding the local oscillator and tuned frequencies. *NOTE 2 Fx is defined in 3.1.18.* *NOTE 3 For outdoor units of home satellite receiving systems highest measured frequency shall be 18 GHz.*	

Where Fx is unknown, the radiated emission measurements shall be performed up to 6 GHz.

5 General Information

5.1 General Description of EUT

Product Name:	Altruist Urban
Model No.:	ES-ALTRUIST-URB-1
Power Supply:	USB input: DC 5V

Disclaimer statement:
The information in this section is provided by the applicant or manufacturer, GTS is not liable to the accuracy, suitability, reliability or/and integrity of the information.

5.2 Test mode and Test voltage

Test mode:	
Operation mode	Keep the EUT in the operation status.
Test voltage:	
AC 230V/50Hz	

5.3 Description of Support Units

Manufacturer	Description	Model	Serial Number
Apple adapter	USB Charger	A1443	N/A

5.4 Monitoring of EUT for All Immunity Test

Visual:	Monitor the indicator light of the EUT.
Audio:	N/A

5.5 Deviation from Standards

None.

5.6 Abnormalities from Standard Conditions

None.

5.7 Test Facility

The test facility is recognized, certified, or accredited by the following organizations:

● **FCC —Registration No.: 381383**

Designation Number: CN5029

Global United Technology Services Co., Ltd., Shenzhen EMC Laboratory has been registered and fully described in a report filed with the (FCC) Federal Communications Commission. The acceptance letter from the FCC is maintained in files.

● **ISED —Registration No.: 9079A**

CAB identifier: CN0091

The 3m Semi-anechoic chamber of Global United Technology Services Co., Ltd. has been registered by Certification and Engineering Bureau of ISED for radio equipment testing.

● **NVLAP (LAB CODE:600179-0)**

Global United Technology Services Co., Ltd., is accredited by the National Voluntary Laboratory Accreditation Program (NVLAP).

5.8 Test Location

All test items were performed at:
Global United Technology Services Co., Ltd. Address: No. 123- 128, Tower A, Jinyuan Business Building, No.2, Laodong Industrial Zone, Xixiang Road, Baoan District, Shenzhen, Guangdong, China 518102 Tel: 0755-27798480; Fax: 0755-27798960

6 Test Instruments List

Radiated Emission:						
Item	**Test Equipment**	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal.Date (mm-dd-yy)**	**Cal.Due date (mm-dd-yy)**
1	3m Semi- Anechoic Chamber	ZhongYu Electron	9.2(L)*6.2(W)* 6.4(H)	GTS250	Apr. 11, 2025	Apr. 10, 2026
2	Control Room	ZhongYu Electron	6.2(L)*2.5(W)* 2.4(H)	GTS251	N/A	N/A
3	EMI Test Receiver	Rohde & Schwarz	ESU26	GTS203	Apr. 12, 2025	Apr. 11, 2026
4	BiConiLog Antenna	SCHWARZBECK MESS-ELEKTRONIK	VULB9168	GTS640	Apr. 12, 2025	Apr. 11, 2026
5	Double -ridged waveguide horn	SCHWARZBECK MESS-ELEKTRONIK	BBHA 9120 D	GTS208	Apr. 11, 2025	Apr. 10, 2026
6	Wideband Radio Communication Tester	Rohde & Schwarz	CMW500	GTS575	Jul. 11, 2025	Jul 10, 2026
s	Loop Antenna	ZHINAN	ZN30900A	GTS534	Nov.22, 2025	Nov.21, 2026
8	Broadband Preamplifier	SCHWARZBECK	BBV9718	GTS535	Apr. 11, 2025	Apr. 10, 2026
9	Amplifier(1GHz-26.5GHz)	HP	8449B	GTS601	Apr. 11, 2025	Apr. 10, 2026
10	Horn Antenna (18GH-40GHz)	Schwarzbeck	BBHA 9170	GTS691	Apr. 11, 2025	Apr. 10, 2026
11	FSV·Signal Analyzer (10Hz-40GHz)	Keysight	FSV-40-N	GTS666	Mar. 11, 2025	Mar. 10, 2026
12	Amplifier	/	LNA-1000-30S	GTS650	Apr. 11, 2025	Apr. 10, 2026
13	CDNE M2+M3-16A	HCT	30MHz-300MHz	GTS692	Nov. 24, 2025	Nov. 23, 2026
14	Wideband Amplifier	/	WDA-01004000-15P35	GTS602	Apr. 11, 2025	Apr. 10, 2026
15	Thermo meter	JINCHUANG	GSP-8A	GTS643	Apr. 15, 2025	Apr. 14, 2026
16	RE cable 1	GTS	N/A	GTS675	Jul. 11, 2025	Jul. 10, 2026
17	RE cable 2	GTS	N/A	GTS676	Jul. 11, 2025	Jul. 10, 2026
18	RE cable 3	GTS	N/A	GTS677	Jul. 11, 2025	Jul. 10, 2026
19	RE cable 4	GTS	N/A	GTS678	Jul. 11, 2025	Jul. 10, 2026
20	RE cable 5	GTS	N/A	GTS679	Jul. 11, 2025	Jul. 10, 2026
21	RE cable 6	GTS	N/A	GTS680	Jul. 11, 2025	Jul. 10, 2026
22	RE cable 7	GTS	N/A	GTS681	Jul. 11, 2025	Jul. 10, 2026
23	RE cable 8	GTS	N/A	GTS682	Jul. 11, 2025	Jul. 10, 2026
24	EMI Test Software	AUDIX	E3-6.100614a	GTS725	N/A	N/A

Conducted Emission						
Item	**Test Equipment**	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal.Date (mm-dd-yy)**	**Cal.Due date (mm-dd-yy)**
1	Shielding Room	ZhongYu Electron	7.3(L)x3.1(W)x2.9(H)	GTS252	Jul. 12, 2022	Jul. 11, 2027
2	EMI Test Receiver	R&S	ESCI 7	GTS552	Apr. 12, 2025	Apr. 11, 2026
3	LISN	ROHDE & SCHWARZ	ENV216	GTS226	Apr. 11, 2025	Apr. 10, 2026
4	Coaxial Cable	GTS	N/A	GTS227	N/A	N/A
5	Thermo meter	JINCHUANG	GSP-8A	GTS642	Apr. 15, 2025	Apr. 14, 2026
6	Absorbing clamp	Elektronik-Feinmechanik	MDS21	GTS229	Apr. 12, 2025	Apr. 11, 2026
7	ISN	SCHWARZBECK	NTFM 8158	GTS565	Apr. 11, 2025	Apr. 10, 2026
8	High voltage probe	SCHWARZBECK	TK9420	GTS537	Apr. 11, 2025	Apr. 10, 2026
9	Antenna end assembly	Weinschel	1870A	GTS560	Apr. 11, 2025	Apr. 10, 2026
10	EMI Test Software	AUDIX	E3-6.100622	GTS726	N/A	N/A
11	Current probe	CYBERTEK	EM5011	GTS698	Jan. 12, 2026	Jan. 11, 2027

ESD						
Item	**Test Equipment**	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal.Date (mm-dd-yy)**	**Cal.Due date (mm-dd-yy)**
1	ESD Simulator	LINCEL	ESD-203B	GTS645	Apr. 12, 2025	Apr. 11, 2026
2	Thermo meter	KTJ	TA328	GTS243	Apr. 15, 2025	Apr. 14, 2026

Conducted Immunity						
Item	**Test Equipment**	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal.Date (mm-dd-yy)**	**Cal.Due date (mm-dd-yy)**
1	Signal Generator	ROHDE & SCHWARZ	SMB 100A	GTS553	Apr. 11, 2025	Apr. 10, 2026
2	CDN	LionCEL	CDN-M3-16	GTS554	Apr. 11, 2025	Apr. 10, 2026
3	CDN	CYBERTEK	EM 5070	GTS559	Apr. 11, 2025	Apr. 10, 2026
4	Power amplifier	rflight	NTWPA-00010475	GTS555	Apr. 11, 2025	Apr. 10, 2026
5	ATT	SUNWAVE	SJ-50-06DB	GTS556	Apr. 11, 2025	Apr. 10, 2026
6	Clamp	SCHAFFNER	KEMZ 801	GTS558	Apr. 12, 2025	Apr. 11, 2026
7	Thermo meter	JINCHUANG	GSP-8A	GTS642	Apr. 15, 2025	Apr. 14, 2026
8	CS Test software	EZ-CS	V3A1	GTS731	N/A	N/A

Harmonic/ Flicker						
Item	**Test Equipment**	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal.Date (mm-dd-yy)**	**Cal.Due date (mm-dd-yy)**
1	Power Analyzer H/F	EMTEST	LFZ-1-16	GTS646	Apr. 11, 2025	Apr. 10, 2026
2	AC POWER SUPPLY	EMTEST	ACS500	GTS236	Apr. 11, 2025	Apr. 10, 2026
3	Thermo meter	JINCHUANG	GSP-8A	GTS639	Apr. 15, 2025	Apr. 14, 2026
4	Harmonics/Flicker Test software	HFA(16A) program	V1.0.0.14	GTS728	N/A	N/A

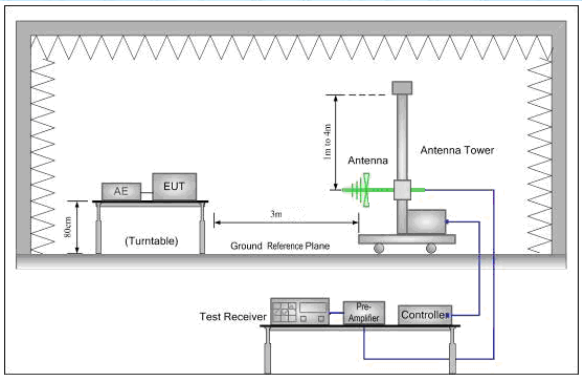
EFT, Surge, Voltage dips and Interruption						
Item	**Test Equipment**	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal.Date (mm-dd-yy)**	**Cal.Due date (mm-dd-yy)**
1	EMTEST system	EMTEST	UCS500N	GTS239	Apr. 11, 2025	Apr. 10, 2026
2	Clamp	EMTEST	HFK	GTS557	Apr. 11, 2025	Apr. 10, 2026
3	Thermo meter	JINCHUANG	GSP-8A	GTS639	Apr. 15, 2025	Apr. 14, 2026
4	EFT/SURGE/DipsTest software	iec control	5.0.9.0	GTS727	N/A	N/A

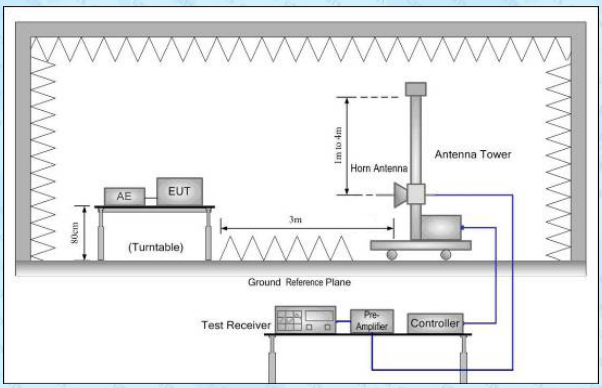
Radiated Immunity						
Item	**Test Equipment**	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal.Date (mm-dd-yy)**	**Cal.Due date (mm-dd-yy)**
1	Probe	STT	SEM-600	GTS648	Apr. 12, 2025	Apr. 11, 2026
2	Stacked Log.-Per.-Broadband Antenna	SCHWARZBECK	STLP 9129	GTS658	Jul. 11, 2025	Jul. 10, 2026
3	MXG vector Signal Generator	Agilent	N5181A	GTS659	Nov. 24, 2025	Nov. 23, 2026
4	Power amplifier	Micotop	MPA-20-1000-250	GTS660	Jul. 11, 2025	Jul. 10, 2026
5	Power amplifier	Micotop	MPA-1000-6000-100	GTS661	Jul. 11, 2025	Jul. 10, 2026
6	EPM SSERIES POWER METER	Agilent	E4419B	GTS662	Nov. 24, 2025	Nov. 23, 2026
7	E-SERIES AVG POWER SENSOR	HP	E9301A	GTS670	Nov. 24, 2025	Nov. 23, 2026
8	Thermo meter	JINCHUANG	GSP-8A	GTS643	Apr. 15, 2025	Apr. 14, 2026
9	RS Test software	EM	V1.2.6	GTS729	N/A	N/A

General used equipment:						
Item	**Test Equipment**	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal.Date (mm-dd-yy)**	**Cal.Due date (mm-dd-yy)**
1	Barometer	KUMAO	SF132	GTS647	Jul. 16, 2025	Jul. 15, 2026

7 Emission Test Results

7.1 Radiated Emission

<table>
<tr><td>Test Requirement:</td><td>EN 55032</td></tr>
<tr><td>Test Method:</td><td>EN 55032</td></tr>
<tr><td>Test Frequency Range:</td><td>30MHz to 6GHz</td></tr>
<tr><td>Class / Severity:</td><td>Class B</td></tr>
<tr><td>Test site:</td><td>Measurement Distance: 3m</td></tr>
<tr><td>Receiver setup:</td><td>
<table>
<tr><td>Frequency</td><td>Detector</td><td>RBW</td><td>VBW</td><td>Value</td></tr>
<tr><td>30MHz-1GHz</td><td>Quasi-peak</td><td>120KHz</td><td>300KHz</td><td>Quasi-peak</td></tr>
<tr><td rowspan="2">Above 1GHz</td><td>Peak</td><td>1MHz</td><td>3MHz</td><td>Peak</td></tr>
<tr><td>AV</td><td>1MHz</td><td>3MHz</td><td>Average</td></tr>
</table>
</td></tr>
<tr><td>Limit:</td><td>
<table>
<tr><td>Frequency</td><td>Limit (dBµV/m @3m)</td><td>Value</td></tr>
<tr><td>30MHz-230MHz</td><td>40.00</td><td>Quasi-peak</td></tr>
<tr><td>230MHz-1GHz</td><td>47.00</td><td>Quasi-peak</td></tr>
<tr><td>1GHz-6GHz</td><td>74.00</td><td>Peak</td></tr>
<tr><td>1GHz-6GHz</td><td>54.00</td><td>Average</td></tr>
</table>
</td></tr>
<tr><td>Test setup:</td><td>Below 1GHz:
 

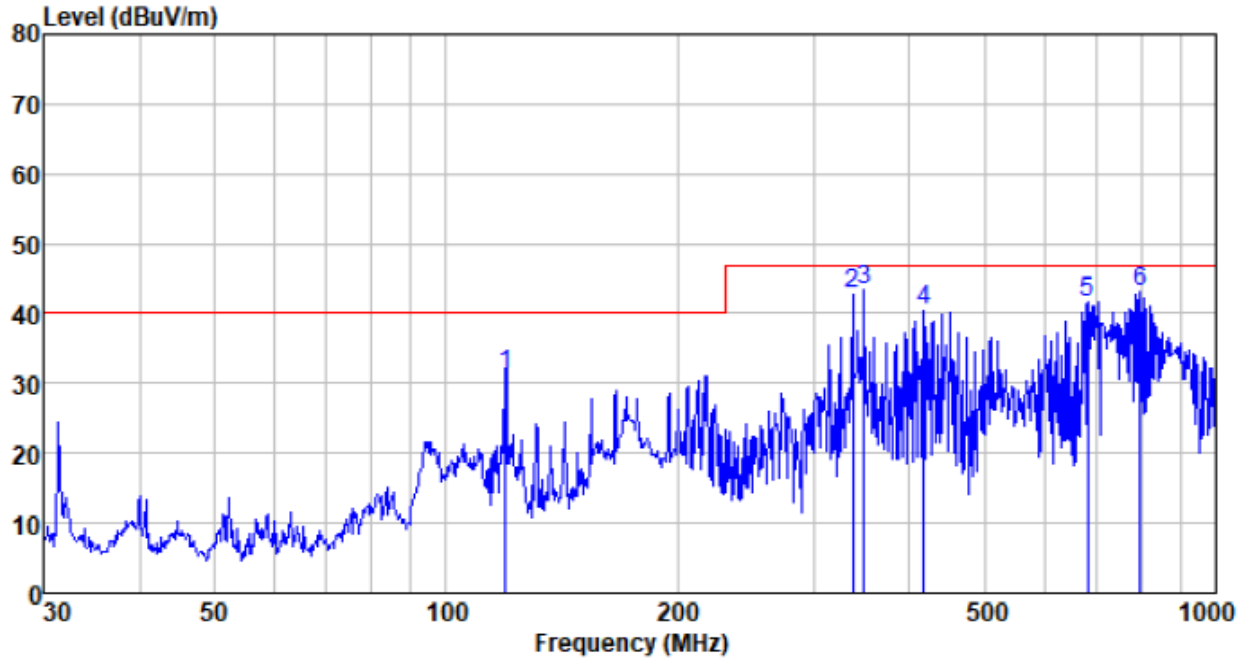
Above 1GHz:
 
</td></tr>
</table>

Test Procedure:	**From 30MHz to 1GHz:** 1. The radiated emissions test was conducted in a semi-anechoic chamber. 2. The tabletop EUT was placed upon a non-metallic table 0.8m above the ground reference plane. And for floor-standing arrangement, the EUT was placed on the horizontal ground reference plane, but separated from metallic contact with the ground reference plane by 0.1m of insulation. 3. Before final measurements of radiated emissions, a pre-scan was performed in the spectrum mode with the peak detector to find out the maximum emissions spectrum plots of the EUT. 4. The frequencies of maximum emission were determined in the final radiated emissions measurement. At each frequency, the EUT was rotated 360°, and the antenna was raised and lowered from 1 to 4 meters in order to determine the maximum disturbance. Measurements were performed for both horizontal and vertical antenna polarization. **Above 1GHz:** 1. The radiated emissions test was conducted in a fully-anechoic chamber. 2. The tabletop EUT was placed upon a non-metallic table 0.8m above the ground reference plane. And for floor-standing arrangement, the EUT was placed on the horizontal ground reference plane, but separated from metallic contact with the ground reference plane by 0.1m of insulation. 3. Before final measurements of radiated emissions, a pre-scan was performed in the spectrum mode with the peak detector to find out the maximum emission spectrum plots of the EUT. 4. The frequencies of maximum emission were determined in the final radiated emissions measurement. At each frequency, the EUT was rotated 360°, and the antenna was raised and lowered from 1 to 4 meters in order to determine the maximum disturbance. Measurements were performed for both horizontal and vertical antenna polarization.					
Test environment:	Temp.:	25 °C	Humid.:	52%	Press.:	1 012mbar
Measurement Record:	Uncertainty: 3.8039dB （30MHz-200MHz） 3.9679dB （200MHz-1GHz） 4.29dB （1GHz-18GHz）					
Test Instruments:	Refer to section 6 for details					
Test mode:	Refer to section 5.2 for details, and only show the worst case.					
Test results:	Pass					

Measurement Data

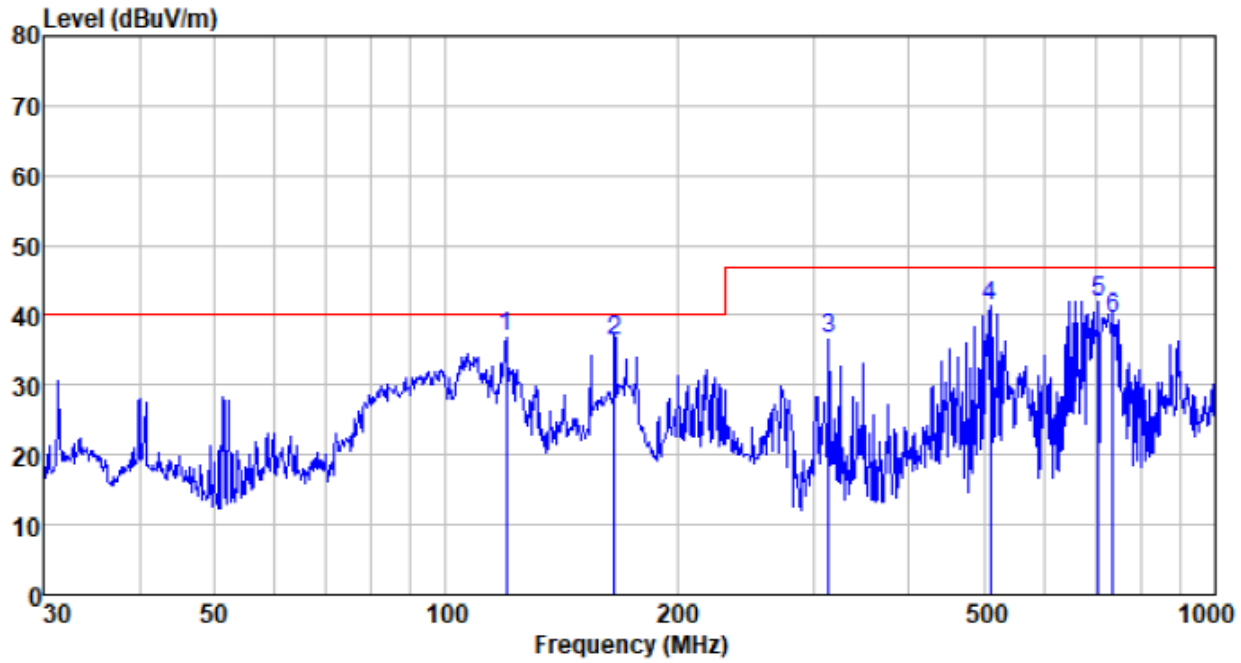
Below 1GHz:

Test mode:	Operation mode	Antenna Polarity:	Horizontal



Freq MHz	Reading level dBuV	Antenna factor dB/m	Cable loss dB	Preamp factor dB	Final level dBuV/m	Limit level dBuV/m	Over limit dB	Remark
119.436	50.16	12.54	2.17	33.55	31.32	40.00	-8.68	QP
337.216	58.78	13.29	3.79	33.13	42.73	47.00	-4.27	QP
349.250	58.98	13.67	3.86	33.11	43.40	47.00	-3.60	QP
417.641	53.98	15.21	4.23	33.01	40.41	47.00	-6.59	QP
682.348	48.36	20.35	5.45	32.44	41.72	47.00	-5.28	QP
796.183	47.37	21.90	6.06	32.21	43.12	47.00	-3.88	QP

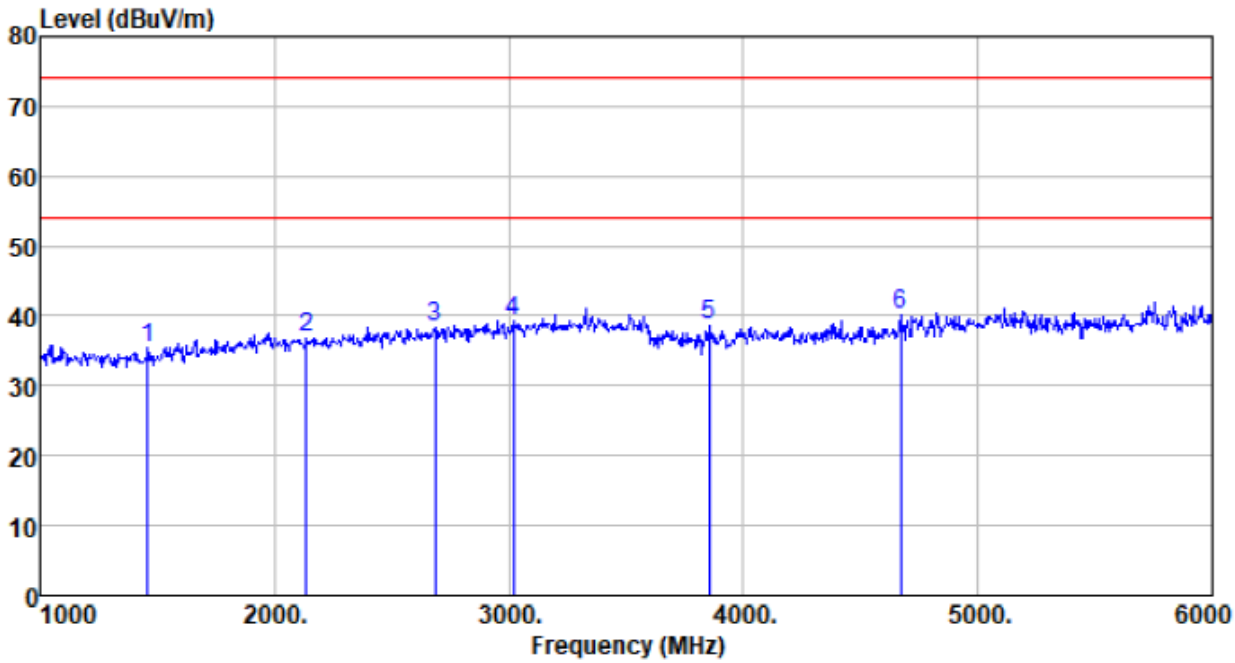
Test mode:	Operation mode	Antenna Polarity:	Vertical



Freq MHz	Reading level dBuV	Antenna factor dB/m	Cable loss dB	Preamp factor dB	Final level dBuV/m	Limit level dBuV/m	Over limit dB	Remark
119.856	55.76	12.59	2.18	33.55	36.98	40.00	-3.02	QP
165.487	53.41	13.75	2.56	33.45	36.27	40.00	-3.73	QP
314.377	53.36	12.68	3.65	33.17	36.52	47.00	-10.48	QP
510.044	52.57	17.00	4.63	32.87	41.33	47.00	-5.67	QP
704.226	48.19	20.58	5.55	32.39	41.93	47.00	-5.07	QP
737.071	44.89	21.14	5.74	32.32	39.45	47.00	-7.55	QP

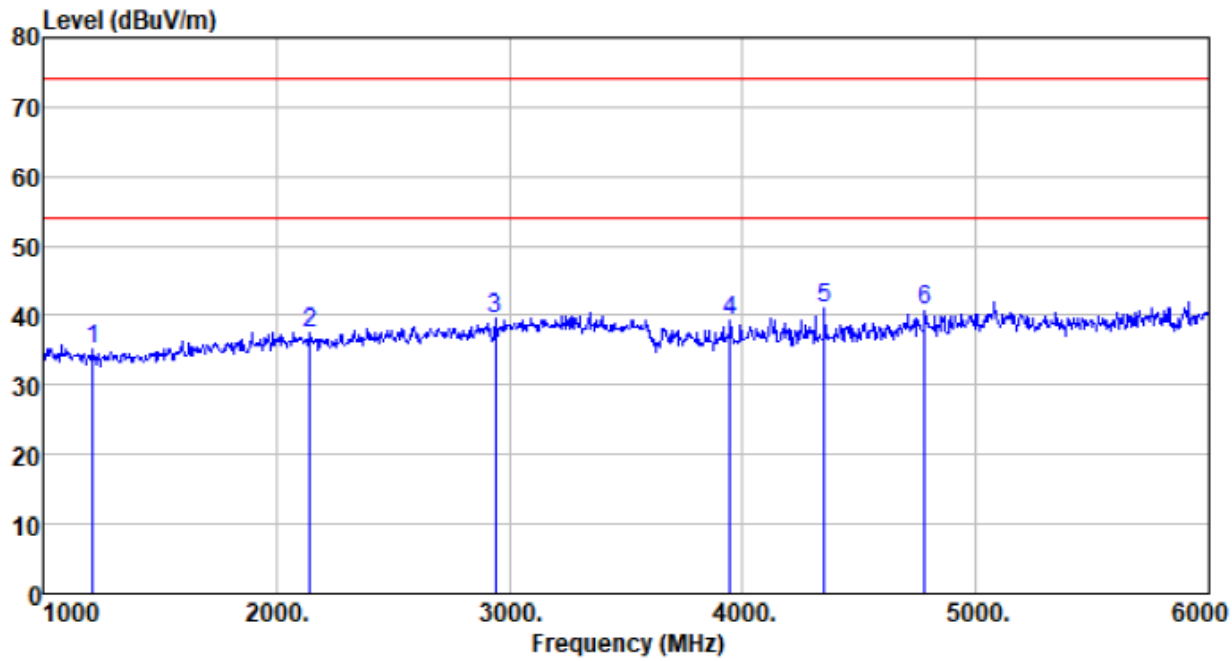
Above 1GHz:

Test mode:	Operation mode	Antenna Polarity:	Horizontal



Freq MHz	Reading level dBuV	Antenna factor dB/m	Cable loss dB	Preamp factor dB	Final level dBuV/m	Limit level dBuV/m	Over limit dB	Remark
1460.000	47.37	24.64	2.26	38.89	35.38	74.00	-38.62	Peak
2135.000	45.78	26.96	2.64	38.45	36.93	74.00	-37.07	Peak
2685.000	45.96	27.84	3.15	38.60	38.35	74.00	-35.65	Peak
3020.000	45.93	28.60	3.44	38.60	39.37	74.00	-34.63	Peak
3855.000	43.82	29.10	3.92	38.29	38.55	74.00	-35.45	Peak
4670.000	43.33	30.68	4.54	38.33	40.22	74.00	-33.78	Peak

Test mode:	Operation mode	Antenna Polarity:	Vertical



Freq MHz	Reading level dBuV	Antenna factor dB/m	Cable loss dB	Preamp factor dB	Final level dBuV/m	Limit level dBuV/m	Over limit dB	Remark
1215.000	47.95	24.31	2.16	39.43	34.99	74.00	-39.01	Peak
2145.000	46.42	26.97	2.64	38.46	37.57	74.00	-36.43	Peak
2940.000	46.25	28.46	3.35	38.60	39.46	74.00	-34.54	Peak
3945.000	44.35	29.22	3.91	38.23	39.25	74.00	-34.75	Peak
4350.000	44.99	29.93	4.26	38.27	40.91	74.00	-33.09	Peak
4780.000	43.57	30.98	4.59	38.36	40.78	74.00	-33.22	Peak

Notes:
The field strength is calculated by adding the Antenna Factor, Cable Factor & Preamplifier. The basic equation with a sample calculation is as follows:
1. Final Level (dBuV/m)=Reading level(dBuV)+Antenna Factor(dB/m)+Cable Loss(dB)-Preamp Factor(dB)
2. If the average limit is met when using a Peak detector, the EUT shall be deemed to meet both peak and average limits. And measurement with the average detector is unnecessary.

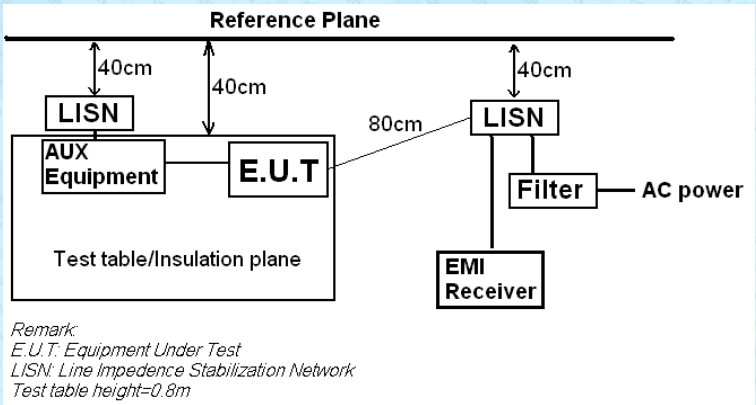
7.2 Conducted Emission

Test Requirement:	EN 55032
Test Method:	EN 55032
Test Frequency Range:	150kHz to 30MHz
Class / Severity:	Class B
Receiver setup:	RBW=9kHz, VBW=30kHz
Limit:	

Frequency range (MHz)	Limit (dBµV)	
	Quasi-peak	Average
0.15-0.5	66 to 56*	56 to 46*
0.5-5	56	46
5-30	60	50

* Decreases with the logarithm of the frequency.

Test setup:



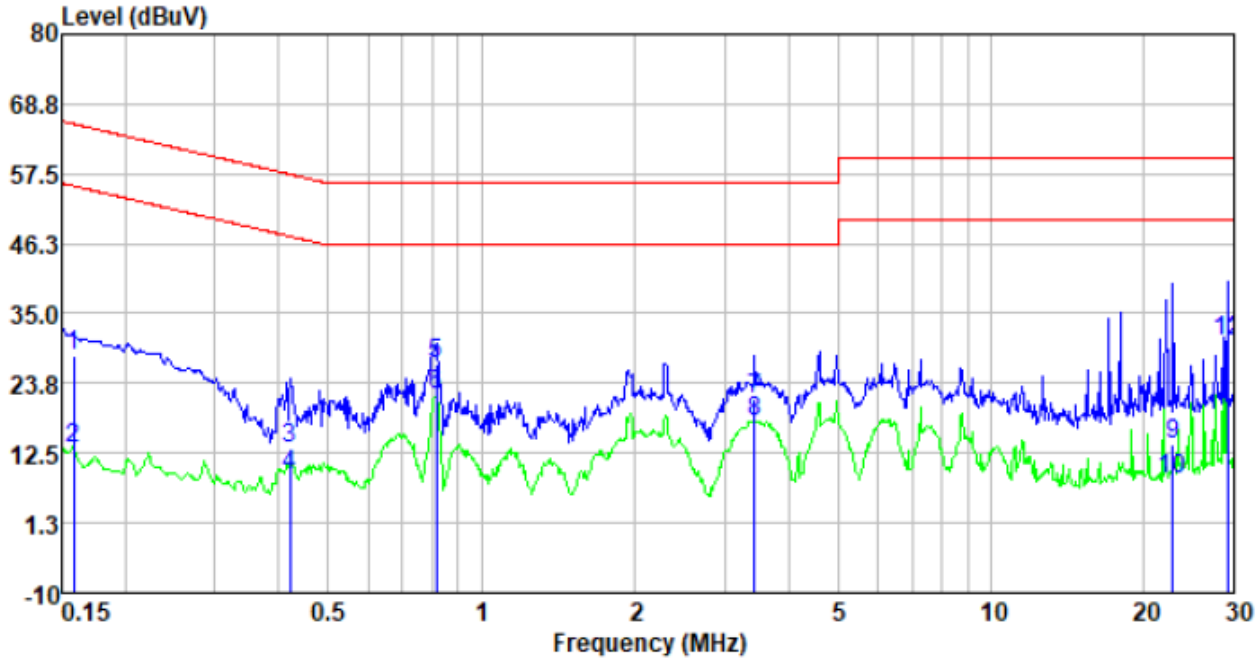
Test procedure:

1. The E.U.T and simulators are connected to the main power through a line impedance stabilization network(LISN). The provide a 50ohm/50uH coupling impedance for the measuring equipment.
2. The peripheral devices are also connected to the main power through a LISN that provides a 50ohm/50uH coupling impedance with 50ohm termination. (Please refers to the block diagram of the test setup and photographs).
3. Both sides of A.C. line are checked for maximum conducted interference. In order to find the maximum emission, the relative positions of equipment and all of the interface cables must be changed according to EN 55032 Class B on conducted measurement.

Test environment:	Temp.:	24 °C	Humid.:	51%	Press.:	1012mbar
Measurement Record:	Uncertainty: 3.44dB					
Test Instruments:	Refer to section 6 for details					
Test mode:	Refer to section 5.2 for details and only show the worst case.					
Test results:	Pass					

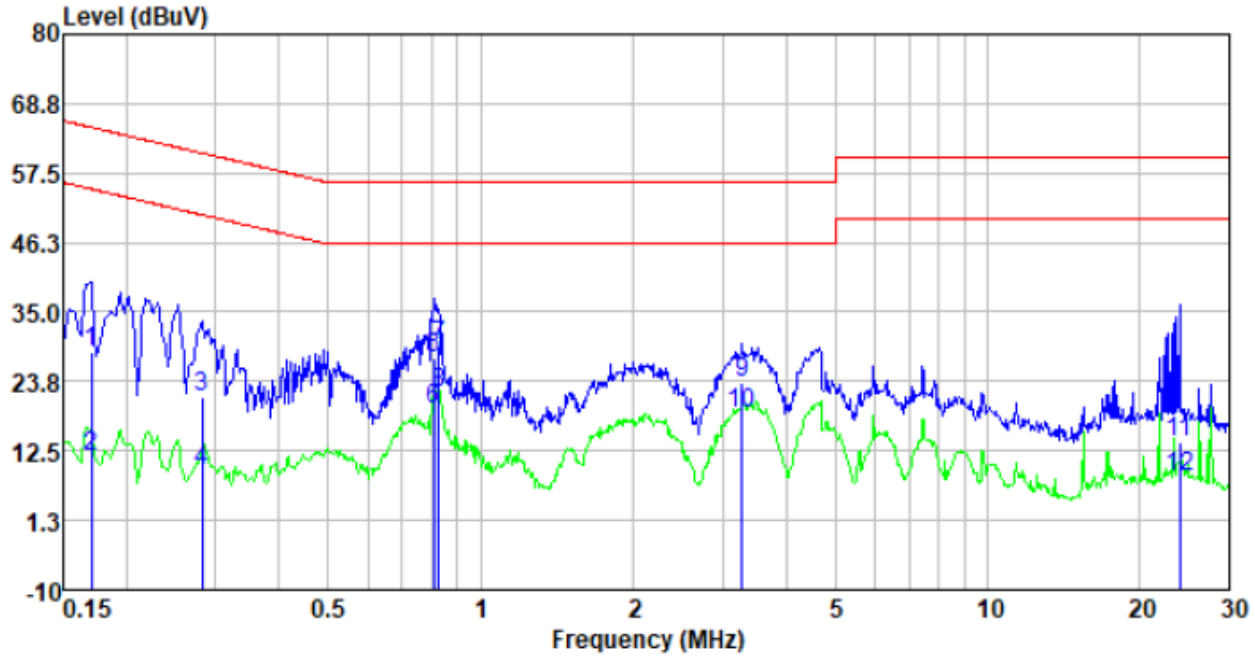
Measurement Data

Test mode:	Operation mode	Phase Polarity:	Line



Freq	Reading level	Correct Factor	Final level	Limit level	Over limit	Remark
MHz	dBuV	dB	dBuV	level	dBuV	
0.16	8.47	19.60	28.07	65.56	-37.49	QP
0.16	-6.50	19.60	13.10	55.56	-42.46	Average
0.42	-6.37	19.60	13.23	57.42	-44.19	QP
0.42	-10.80	19.60	8.80	47.42	-38.62	Average
0.81	7.21	19.61	26.82	56.00	-29.18	QP
0.81	2.25	19.61	21.86	46.00	-24.14	Average
3.44	1.66	19.67	21.33	56.00	-34.67	QP
3.44	-2.05	19.67	17.62	46.00	-28.38	Average
22.78	-6.02	20.03	14.01	60.00	-45.99	QP
22.78	-11.64	20.03	8.39	50.00	-41.61	Average
29.09	10.12	20.34	30.46	60.00	-29.54	QP
29.09	10.02	20.34	30.36	50.00	-19.64	Average

Test mode:	Operation mode	Phase Polarity:	Neutral



Freq	Reading level	Correct Factor	Final level	Limit level	Over limit	Remark
MHz	dBuV	dB	dBuV	level	dBuV	
0.17	8.82	19.60	28.42	64.94	-36.52	QP
0.17	-7.85	19.60	11.75	54.94	-43.19	Average
0.28	1.54	19.60	21.14	60.76	-39.62	QP
0.28	-10.37	19.60	9.23	50.76	-41.53	Average
0.81	8.04	19.61	27.65	56.00	-28.35	QP
0.81	-0.48	19.61	19.13	46.00	-26.87	Average
0.83	9.76	19.61	29.37	56.00	-26.63	QP
0.83	2.19	19.61	21.80	46.00	-24.20	Average
3.28	3.84	19.67	23.51	56.00	-32.49	QP
3.28	-1.01	19.67	18.66	46.00	-27.34	Average
23.89	-6.17	20.07	13.90	60.00	-46.10	QP
23.89	-11.93	20.07	8.14	50.00	-41.86	Average

Notes:
1. An initial pre-scan was performed on the line and neutral lines with peak detector.
Quasi-Peak and Average measurement were performed at the frequencies with maximized peak emission
2. The disturbance voltage is calculated by adding the LISN Factor & Cable Loss Factor. The basic equation with a sample calculation is as follows:
Correct Factor(dB)=LISN Factor(dB)+Cable Loss(dB)
Final level(dBuV)=Reading level(dBuV)+Correct Factor(dB)

7.3 Harmonic Emission

Test Requirement:	EN IEC 61000-3-2
Test Method:	N/A: See Remark Below
Remark	There is no need for Harmonics test to be performed on this product (rated power is less than 75W) in accordance with EN IEC 61000-3-2. For further details, please refer to Clause 7, Note 1 of EN IEC 61000-3-2 which states: "For the following categories of equipment limits are not specified in this edition of the standard. Note 1: Equipment with a rated power of 75W or less, other than lighting equipment."

7.4 Flicker Emission

Test Requirement:	EN 61000-3-3					
Test Method:	EN 61000-3-3					
Class/Severity:	Clause 5 of EN 61000-3-3					
Measurement Time:	10 min					
Detector:	As per EN 61000-3-3					
Test environment:	Temp.:	24 °C	Humid.:	51%	Press.:	1 012mbar
Test Instruments:	Refer to section 6 for details					
Test mode:	Refer to section 5.2 for details					
Test results:	Pass					

Measurement Data

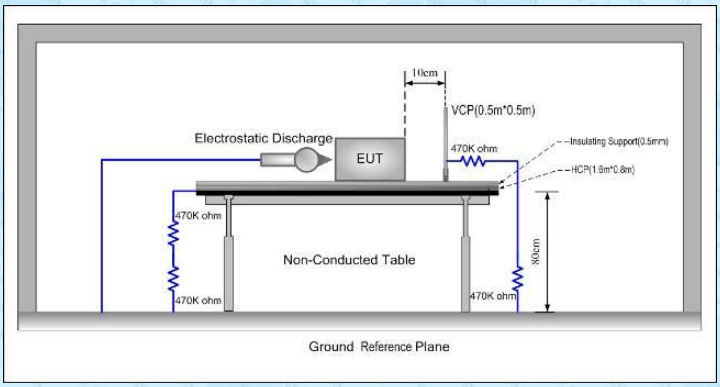
	EUT values	Limit	Result
Pst	0.028	1.00	PASS
Plt	0.028	0.65	PASS
dc [%]	0.000	3.30	PASS
dmax [%]	0.064	4.00	PASS
dt [s]	0.000	0.50	PASS

8 Immunity Test Results

8.1 Performance Criteria Description in Clause 8 of EN 55035

Criterion A:	The equipment shall continue to operate as intended without operator intervention. No degradation of performance, loss of function or change of operating state is allowed below a performance level specified by the manufacturer when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.
Criterion B:	During the application of the disturbance, degradation of performance is allowed. However, no unintended change of actual operating state or stored data is allowed to persist after the test. After the test, the equipment shall continue to operate as intended without operator intervention; no degradation of performance or loss of function is allowed, below a performance level specified by the manufacturer, when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level (or the permissible performance loss), or recovery time, is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.
Criterion C:	Loss of function is allowed, provided the function is self-recoverable, or can be restored by the operation of the controls by the user in accordance with the manufacturer's instructions. A reboot or re-start operation is allowed. Information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

8.2 Electrostatic discharge

Test Requirement:	EN 55035
Test Method:	EN 61000-4-2
Discharge Voltage:	Contact Discharge:±4kV Air Discharge: ±2kV, ±4kV, ±8kV HCP/VCP:±4kV
Polarity:	Positive & Negative
Number of Discharge:	Contact Discharge: Minimum 10 times at each test point, Air Discharge: Minimum 10 times at each test point.
Discharge Mode:	Single Discharge
Discharge Period:	1 second minimum
Performance Criterion:	Criterion B
Test setup:	
Test Procedure:	**1. Air discharge:** The test was applied on non-conductive surfaces of EUT. The round discharge tip of the discharge electrode was approached as fast as possible to touch the EUT. After each discharge, the discharge electrode was removed from the EUT. The generator was re-triggered for a new single discharge and repeated 10 times for each pre-selected test point. This procedure was repeated until all the air discharge completed **2. Contact Discharge:** The test was applied on conductive surfaces of EUT. the generator was re-triggered for a new single discharge and repeated 25 times for each pre-selected test point. the tip of the discharge electrode was touch the EUT before the discharge switch was operated. **3. Indirect discharge for horizontal coupling plane** At least 10 single discharges shall be applied at the front edge of each HCP opposite the centre point of each unit of the EUT and 0.1m from the front of the EUT. The long axis of the discharge electrode shall be in the plane of the HCP and perpendicular to its front edge during the discharge. Consideration should be given to exposing all sides of the EUT. **4. Indirect discharge for vertical coupling plane** At least 10 single discharges were applied to the center of one vertical edge of the coupling plane. The coupling plane, of dimensions 0.5m X 0.5m, was placed parallel to, and positioned at a distance of 0.1m from the EUT. Discharges were applied to the coupling plane, with this plane in sufficient different positions that the four faces of the EUT are completely illuminated.

Test environment:	Temp.:	24 °C	Humid.:	51%	Press.:	1 012mbar
Test Instruments:	Refer to section 6 for details					
Test mode:	Refer to section 5.2 for details					
Test results:	Pass					

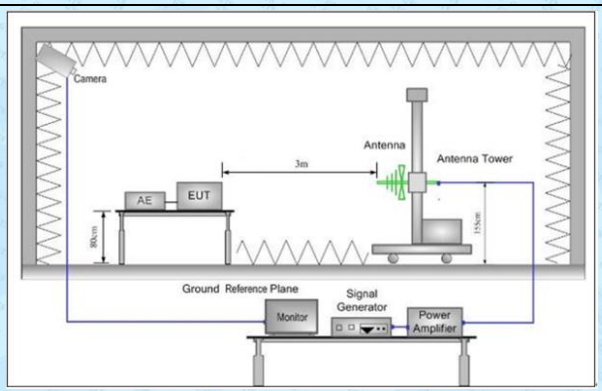
Measurement Record:

Test points:	I: N/A II: All plastic seams			
Direct discharge				
Discharge Voltage (KV)	**Type of discharge**	**Test points**	**Observations Performance**	**Result**
± 4	Contact	I	N/A	N/A
± 2, ± 4, ± 8	Air	II	A	Pass
Indirect discharge				
Discharge Voltage (KV)	**Type of discharge**	**Test points**	**Observation Performance**	**Result**
± 4	HCP-Bottom/Top/ Front/Back/Left/Right	Edge of the HCP	A	Pass
± 4	VCP-Bottom/Top/ Front/Back/Left/Right	Center of the VCP	A	Pass

Remark:

Performance Criteria: A, B, C: Refer to section 8.1 for details

8.3 Continuous RF electromagnetic field disturbances

Test Requirement:	EN 55035
Test Method:	EN 61000-4-3
Frequency range:	80MHz to 5GHz
Test Level:	3V/m
Modulation:	80%, 1kHz Amplitude Modulation
Performance Criterion:	Criterion A
Test setup:	
Test Procedure:	1. For table-top equipment, the EUT was placed in the chamber on a non-conductive table 0.8m high. For arrangement of floor-standing equipment, the EUT was mounted on a non-conductive support 0.1m above the supporting plane. For human body-mounted equipment, the EUT may be tested in the same manner as table top items. 2. If possible, a minimum of 1 m of cable is exposed to the electromagnetic field. Excess length of cables interconnecting units of the EUT shall be bundled low-inductively in the approximate center of the cable to form a bundle 30 cm to 40 cm in length. 3. The EUT was initially placed with one face coincident with the calibration plane. The EUT face being illuminated was contained within the UFA (Uniform Field Area). 4. The frequency ranges to be considered were swept with the signal modulated and pausing to adjust the RF signal level or to switch oscillators and antennas as necessary.Where the frequency range was swept incrementally, the step size was not exceed 1 % of the preceding frequency value. 5. The dwell time of the amplitude modulated carrier at each frequency was not be less than the time necessary for the EUT to be exercised and to respond, and was not less than 0,5 s. 6. The test normally was performed with the generating antenna facing each side of the EUT. 7. The polarization of the field generated by each antenna necessitates testing each selected side twice, once with the antenna positioned vertically and again with the antenna positioned horizontally. 8. The EUT was performed in a configuration to actual installation conditions, a video camera and/or a audio monitor were used to monitor the performance of the EUT.
Test environment:	Temp.: 25 °C | Humid.: 52% | Press.: 1 012mbar
Test Instruments:	Refer to section 6 for details
Test mode:	Refer to section 5.2 for details

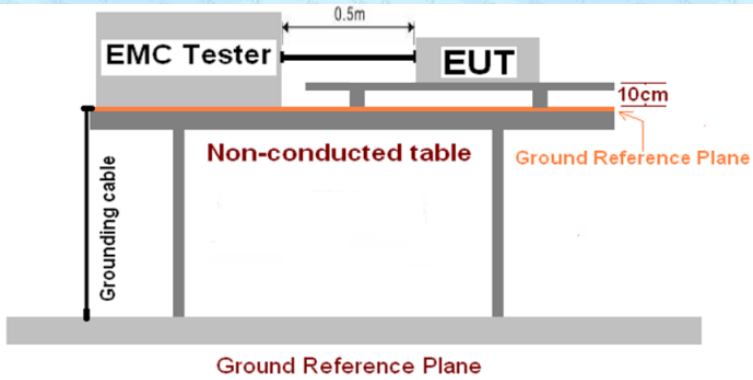
Test results:	Pass

Measurement Record:

Frequency	Level	Modulation	Antenna Polarization	EUT Face	Observations (Performance Criterion)	Result
80 MHz-1 GHz	3 V/m	1 kHz, 80 % Amp. Mod, 1 % increment, dwell time=3seconds	V	Front	A	Pass
			H		A	Pass
			V	Rear	A	Pass
			H		A	Pass
			V	Left	A	Pass
			H		A	Pass
			V	Right	A	Pass
			H		A	Pass
			V	Top	A	Pass
			H		A	Pass
			V	Bottom	A	Pass
			H		A	Pass
1800MHz, 2600MHz 3500MHz 5000MHz	3 V/m	1 kHz, 80 % Amp. Mod, 1 % increment, dwell time=2seconds	V	Front	A	Pass
			H		A	Pass
			V	Rear	A	Pass
			H		A	Pass
			V	Left	A	Pass
			H		A	Pass
			V	Right	A	Pass
			H		A	Pass
			V	Top	A	Pass
			H		A	Pass
			V	Bottom	A	Pass
			H		A	Pass

Remark: Performance Criteria: A, B, C: Refer to section 8.1 for details

8.4 Electrical fast transients

Test Requirement:	EN 55035					
Test Method:	EN 61000-4-4					
Test Level:	1.0kV					
Polarity:	Positive & Negative					
Test signal specification:	Rise time=5ns, Duration time=50ns; Burst Duration=15ms, Burst Period=300ms; Repetition Frequency=5KHz					
Test Duration:	2 minute per level & polarity					
Performance Criterion:	Criterion B					
Test setup:						
Test Procedure:	1. The EUT and its simulators were placed on the ground reference plane and were insulated from it by a wood support 0.1m + 0.01m thick. The ground reference plane was 1m*1m metallic sheet with 0.65mm minimum thickness. 2. This reference ground plane was project beyond the EUT by at least 0.1m on all sides and the minimum distance between EUT and all other conductive structure, except the ground plane was more than 0.5m. 3. All cables to the EUT was placed on the wood support, cables not subject to EFT/B was routed as far as possible from the cable under test to minimize the coupling between the cables. 4. The length of power lines between the coupling device and the EUT is 0.5m 5. The EUT is connected to the power mains through a coupling device that directly couples the EFT/B interference signal. 6. Each of the Line and Neutral conductors is impressed with burst noise for 2 minutes.					
Test environment:	Temp.:	26 °C	Humid.:	54%	Press.:	1 012mbar
Test Instruments:	Refer to section 6 for details					
Test mode:	Refer to section 5.2 for details					
Test results:	Pass					

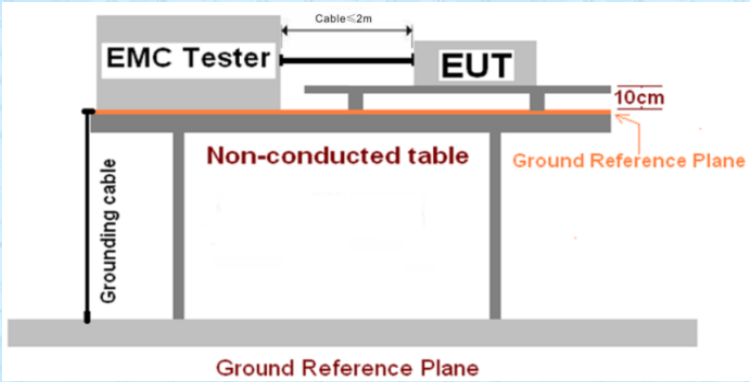
Measurement Record:

Lead under Test	Level (±kV)	Coupling Direct/Clamp	Observations (Performance Criterion)	Result
L	± 1.0	Direct	A	Pass
N	± 1.0	Direct	A	Pass
L-N	± 1.0	Direct	A	Pass

Remark:

Performance Criteria: A, B, C: Refer to section 8.1 for details

8.5 Surges

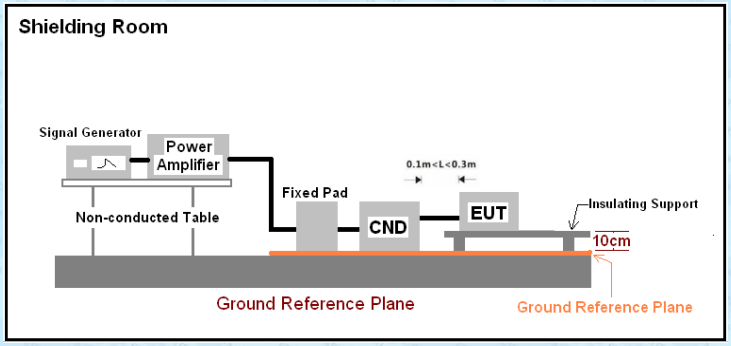
Test Requirement:	EN 55035
Test Method:	EN 61000-4-5
Test Level:	0.5kV,1kV line to line: Differential mode 0.5kV,1kV,2kV line to earth: Common mode
Polarity:	Positive & Negative
Generator source impedance:	2Ω (line-line coupling) 12Ω (line-earth coupling)
Test signal specification:	Rise time=1.2us, Duration time=50us; Test Interval: 60s between each surge;
No. of surges:	Five positive pulses line-to-neutral at 90° phase Five negative pulses line-to-neutral at 270° phase Five positive pulses line-to-earth at 90° phase Five negative pulses line-to-earth at 270° phase Five negative pulses neutral-to-earth at 90° phase Five positive pulses neutral-to-earth at 270° phase
Performance Criterion:	Criterion B
Test setup:	
Test Procedure:	1. For line-to-line coupling mode, provide a 1kV 1.2/50us voltage surge (at open-circuit condition) and 8/20us current surge to EUT selected points, and for active line / neutral lines to ground are same except test level is 2kV. 2. At least 5 positive and 5 negative (polarity) tests with a maximum 1/min repetition rate are applied during test. 3. Different phase angles are done individually. 4. Record the EUT operating situation during compliance test and decide the EUT immunity criterion for above each test.
Test environment:	Temp.: 26 °C | Humid.: 53% | Press.: 1 012mbar
Test Instruments:	Refer to section 6 for details
Test mode:	Refer to section 5.2 for details
Test results:	Pass

Measurement Record:

Location	Level(kV)	Pulse No	Surge Interval	Phase (deg)	Observations (Performance Criterion)	Result
L-N	+0.5, +1	5	60s	90°	A	Pass
	-0.5, -1			270°	A	Pass

Remark:
Performance Criteria: A, B, C: Refer to section 8.1 for details.

8.6 Continuous induced RF disturbance

Test Requirement:	EN 55035
Test Method:	EN 61000-4-6
Frequency range:	0.15MHz to 10MHz:3V 10MHz to 30MHz:3V to 1V 30MHz to 80MHz:1V
Test Level:	3V rms on AC Ports
Performance Criterion:	Criterion A
Test setup:	
Test Procedure:	1. The EUT are placed on an insulating support 0.1m high above a ground reference plane. CDN (coupling and decoupling device) is placed on the ground plane about 0.3m from EUT. Cables between CDN and EUT are as short as possible, and their height above the ground reference plane shall be between 30 and 50 mm (where possible). 2. The disturbance signal described below is injected to EUT through CDN. 3. The EUT operates within its operational mode(s) under intended climatic conditions after power on. 4. Recording the EUT operating situation during compliance testing and decide the EUT immunity criterion.
Test environment:	Temp.: 24 °C Humid.: 51% Press.: 1 012mbar
Test Instruments:	Refer to section 6 for details
Test mode:	Refer to section 5.2 for details
Test results:	Pass

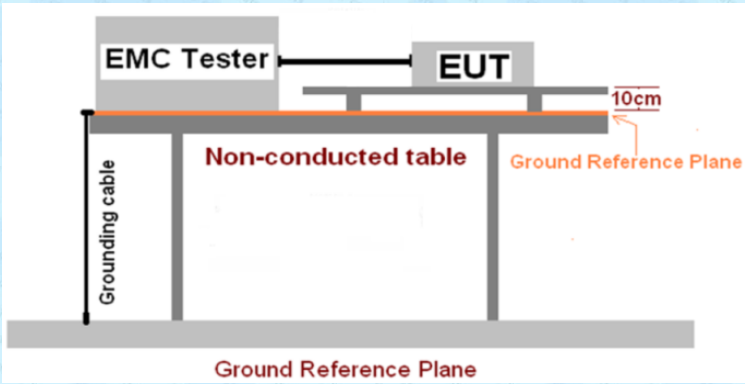
Measurement Record:

Level	Injected Position	Modulation	Observations (Performance Criterion)	Result
3(0.15MHz-10MHz)	CDN	1 kHz, 80 % Amp. Mod, 1 % increment, dwell time=2seconds	A	Pass
3 to 1 (10MHz-30MHz, Lines)				
1 (30MHz-80MHz)				

Remark:

Performance Criteria: A, B, C: Refer to section 8.1 for details.

8.7 Voltage dips and Voltage interruptions

Test Requirement:	EN 55035
Test Method:	EN 61000-4-11
Test Level:	0% of VT(Supply Voltage) for 0.5 period 70% of VT(Supply Voltage) for 25 period 0% of VT(Supply Voltage) for 250 period
Number of Dips / Interruptions:	3 per Level
Performance Criterion:	>95% VD, 0.5 period----Performance criterion: B 30% VD, 25 period----Performance criterion: C >95% VI, 250 period----Performance criterion: C
Test setup:	
Test Procedure:	1. The EUT and test generator were setup as shown on above setup photo. 2. The interruptions are introduced at selected phase angles with specified duration. 3. Record any degradation of performance.
Test environment:	Temp.: 26 °C Humid.: 53% Press.: 1 012mbar
Test Instruments:	Refer to section 6 for details
Test mode:	Refer to section 5.2 for details
Test results:	Pass

Measurement Record:

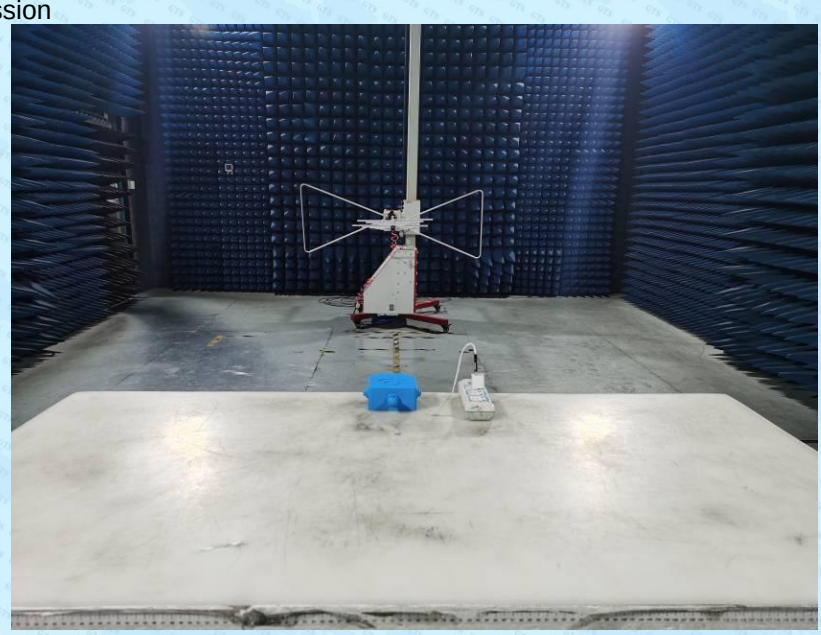
Test Level % U_T	Duration (Periods)	Phase angle	No. of drop out	Time between dropout	Observations (Performance Criterion)	Result
0	0.5	0°, 90°, 180°, 270°	3	10s	A	Pass
70	25	0°, 90°, 180°, 270°	3	10s	A	Pass
0	250	0°, 90°, 180°, 270°	3	10s	C	Pass

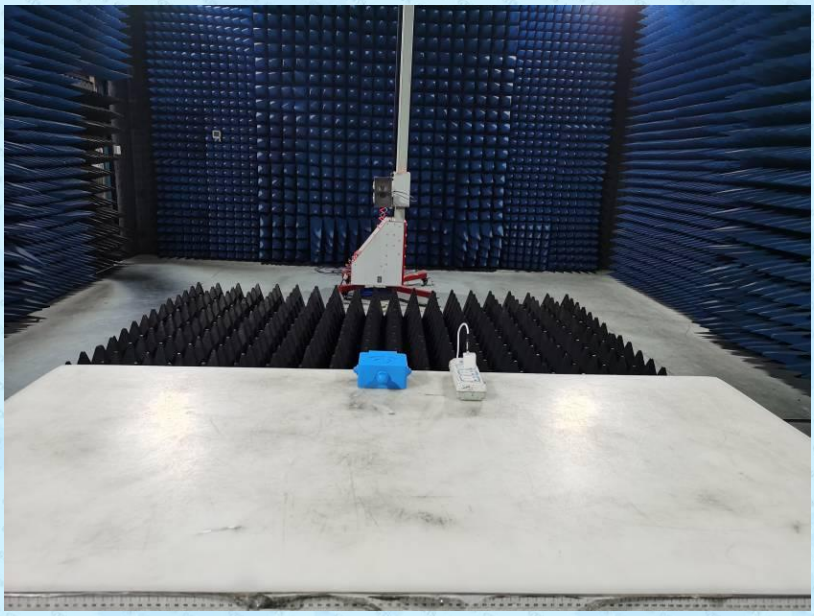
Remark:

Performance Criteria: A, B, C: Refer to section 8.1 for details.

9 Test Setup Photo

Radiated Emission

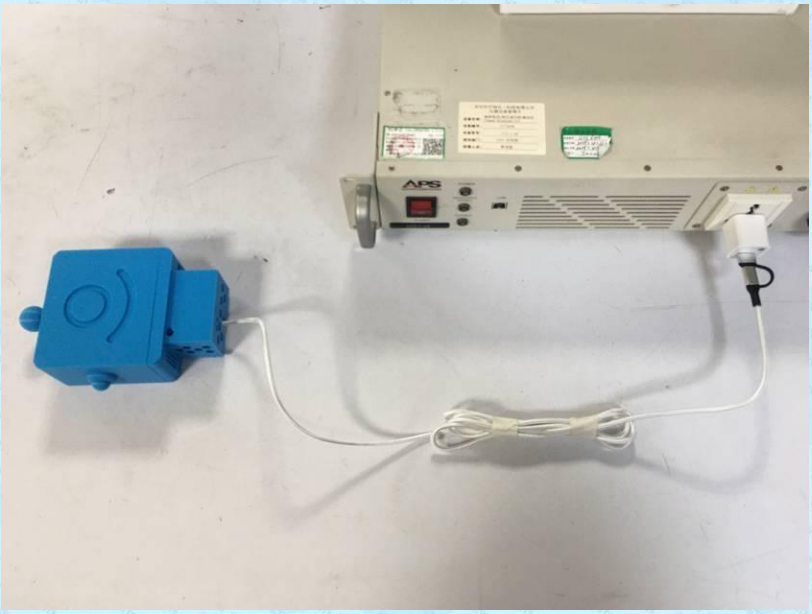




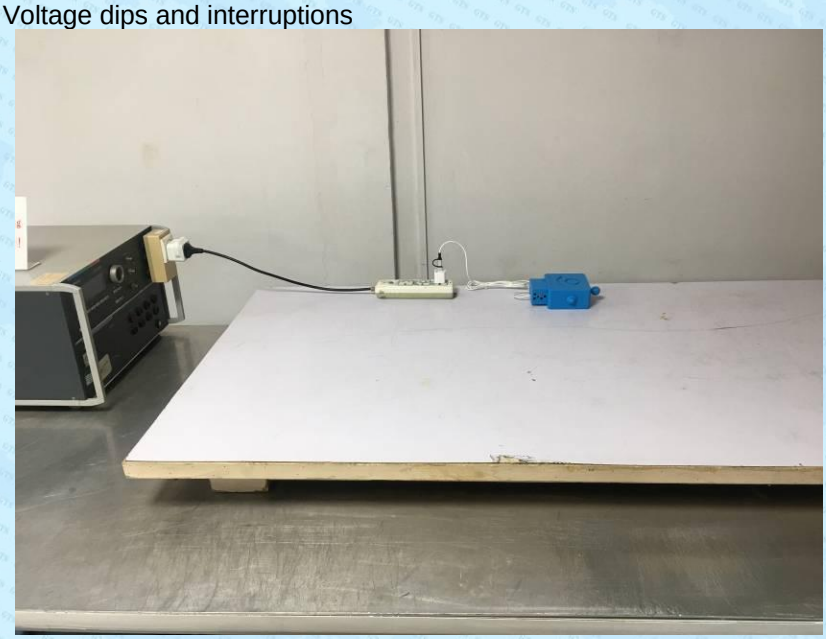
Conducted Emission

Electrostatic discharge

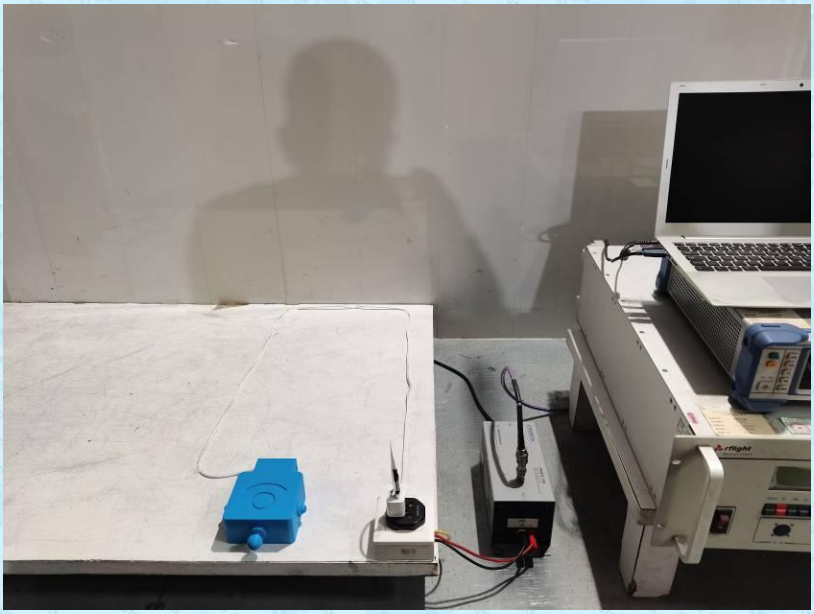
Harmonic/Flicker



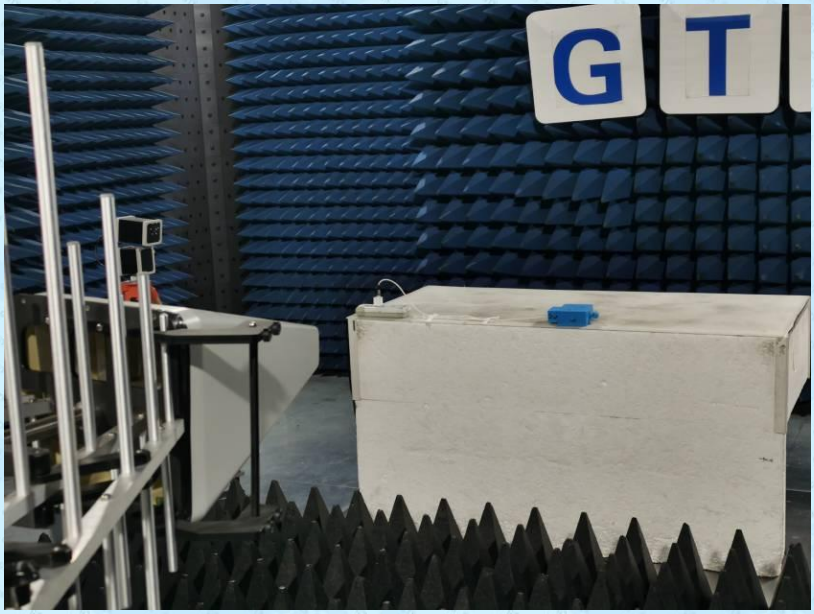
EFT/ Surges/ Voltage dips and interruptions



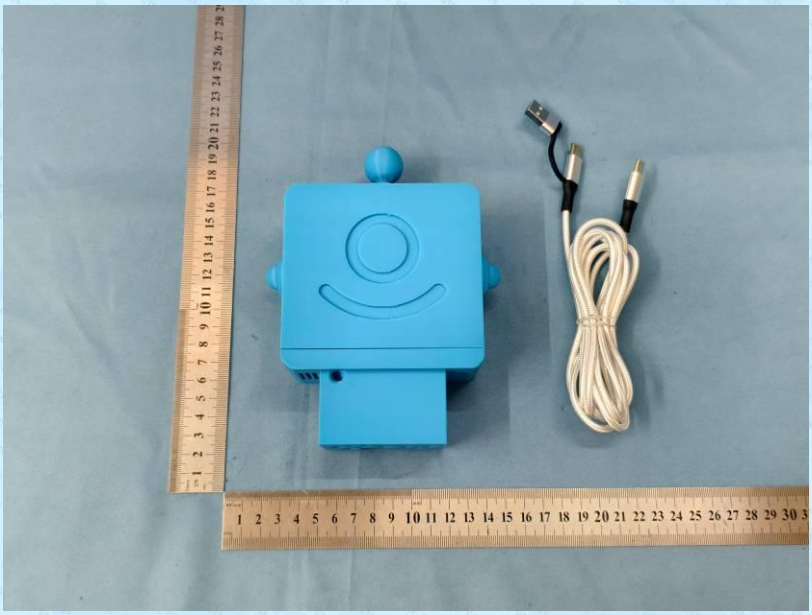
CS

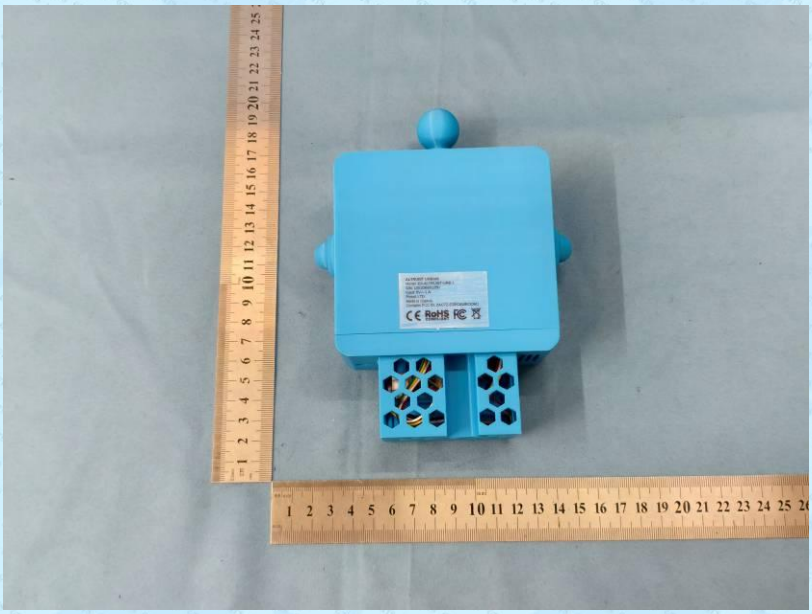


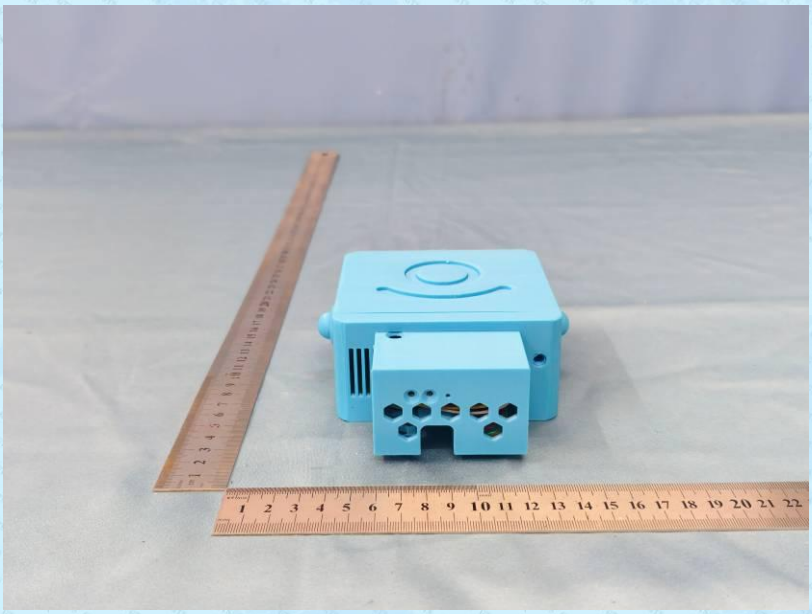
RS

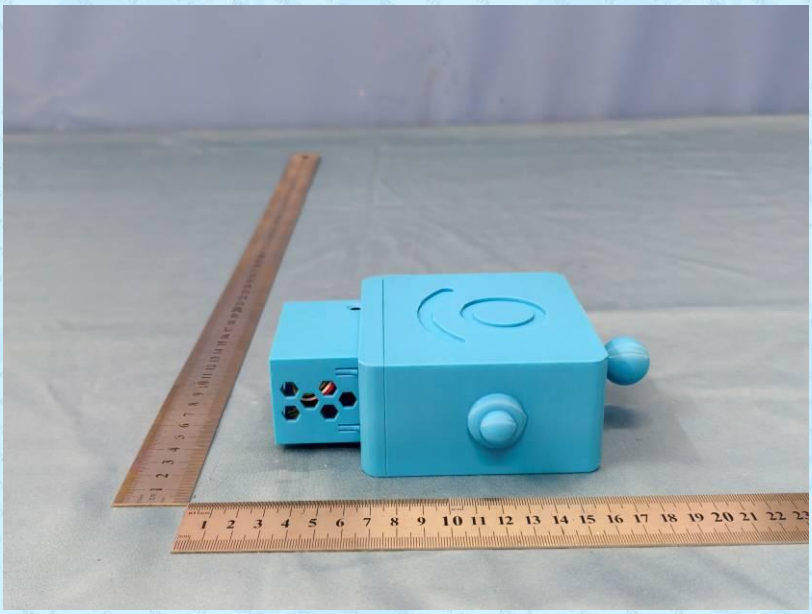


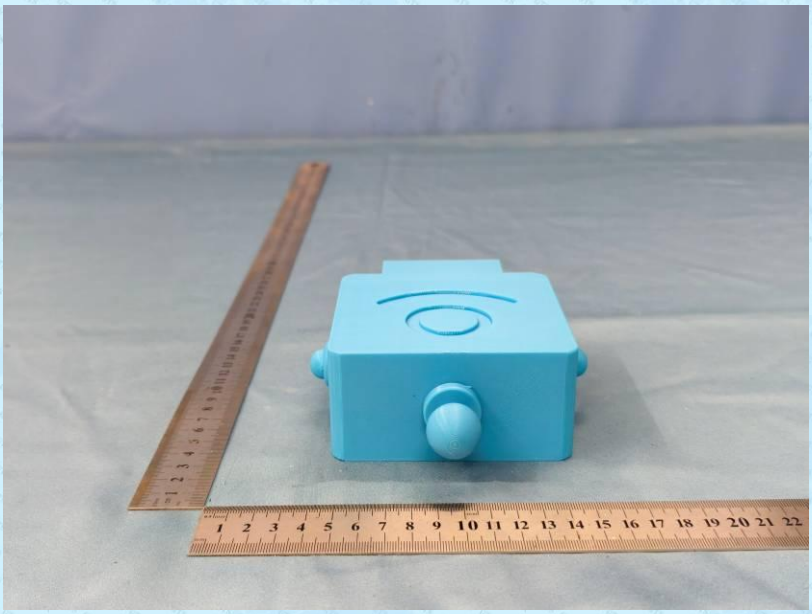
10 EUT Constructional Details

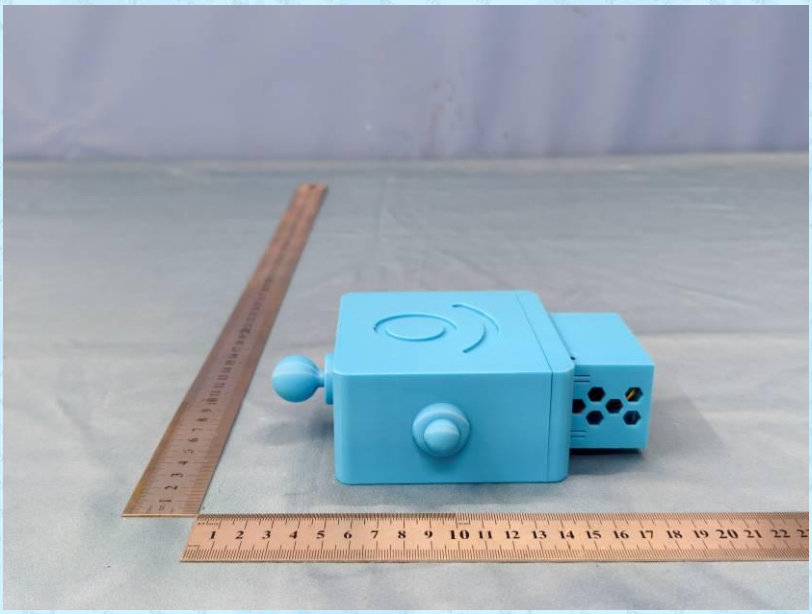


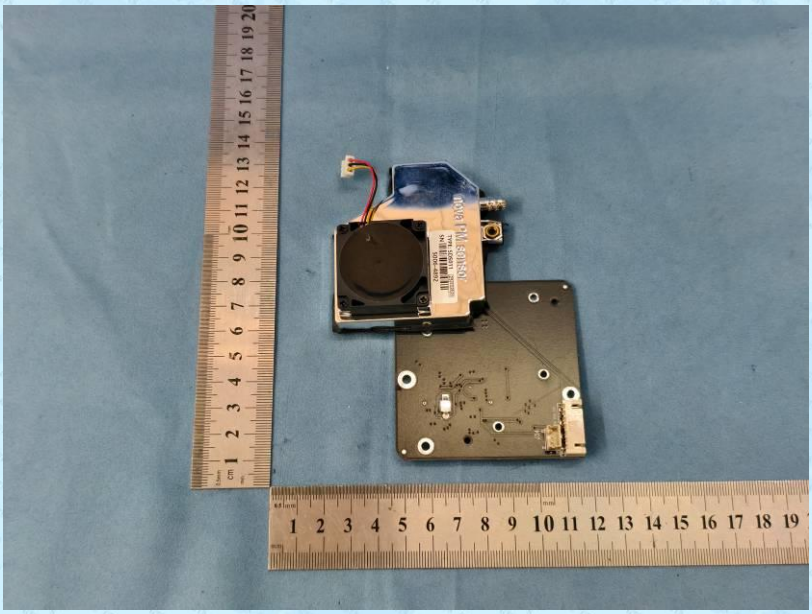


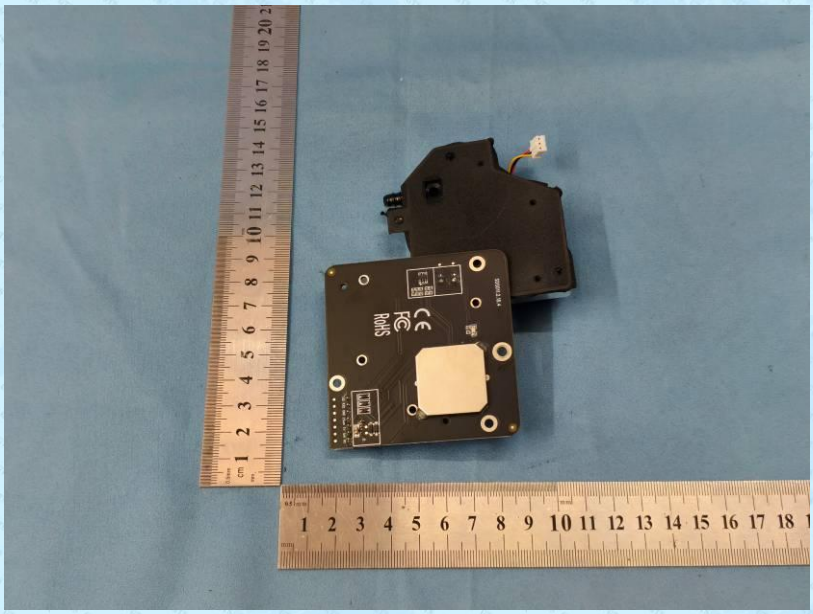


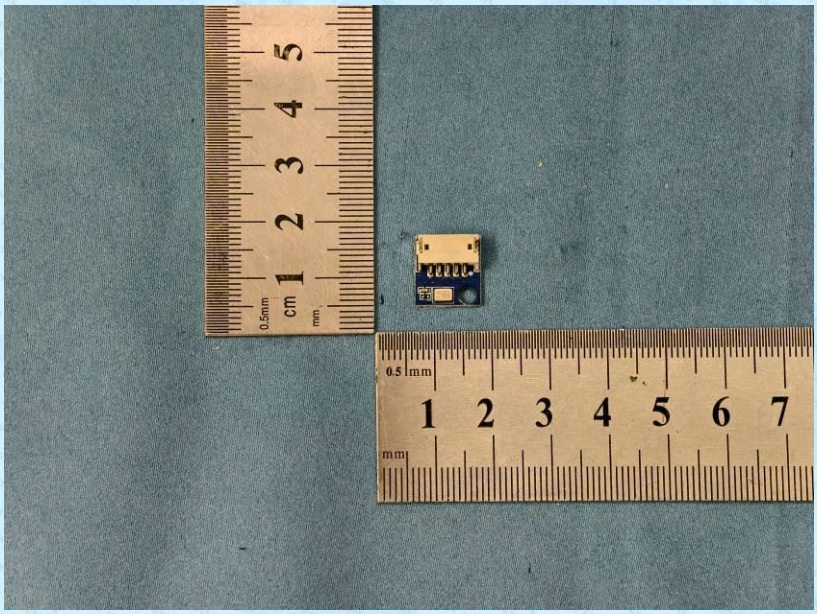


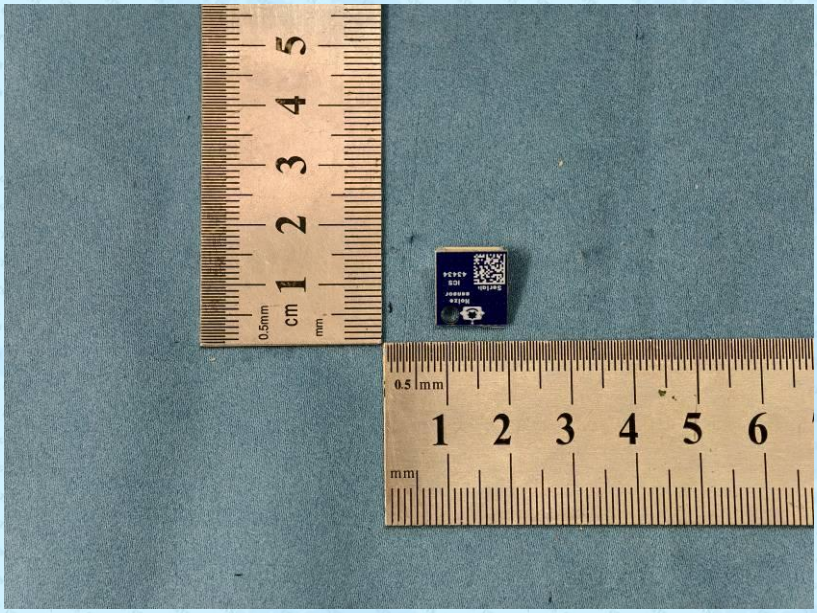


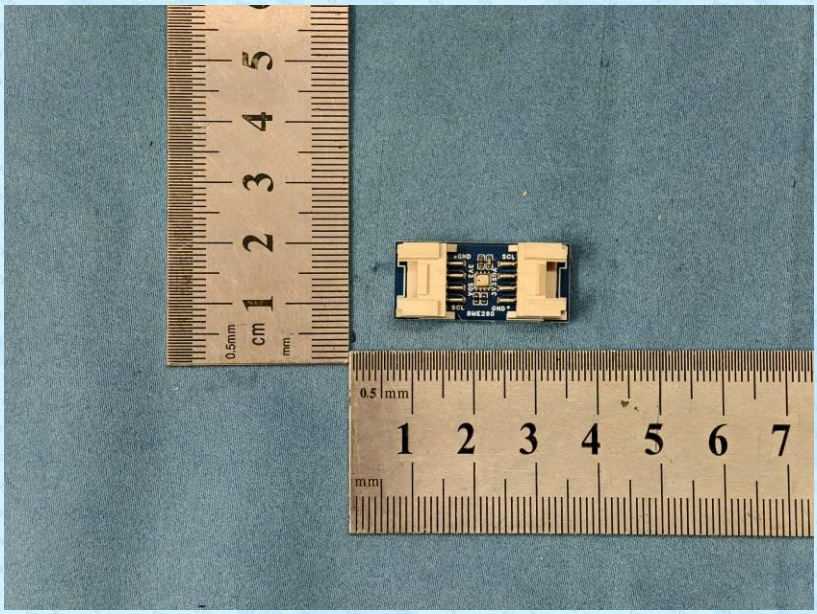


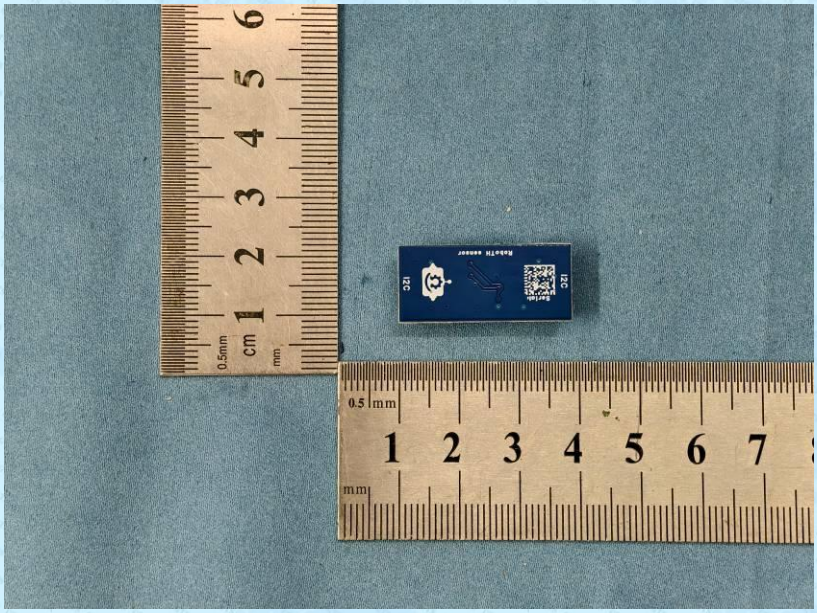


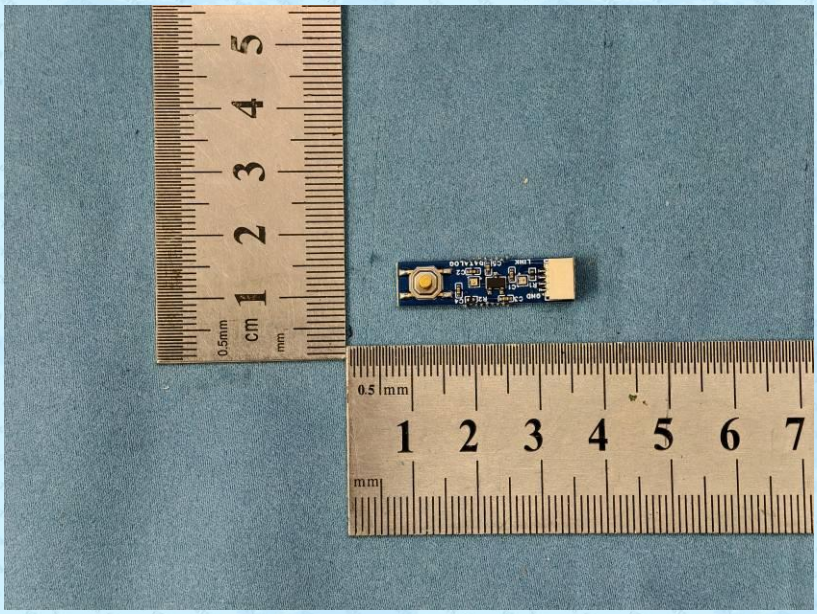


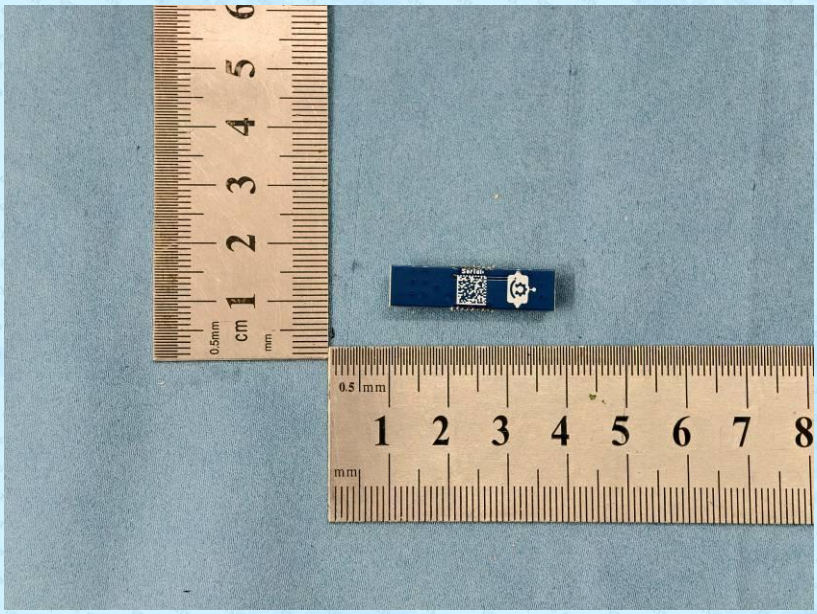


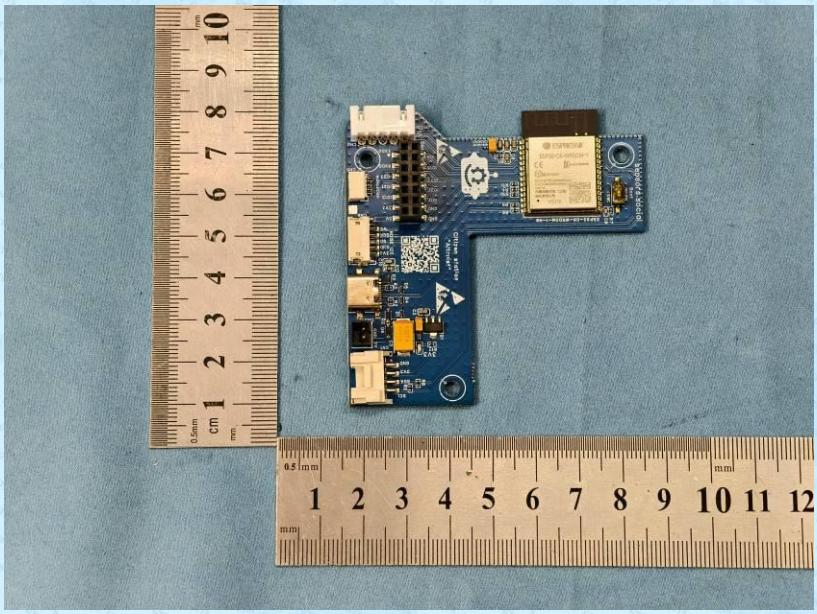


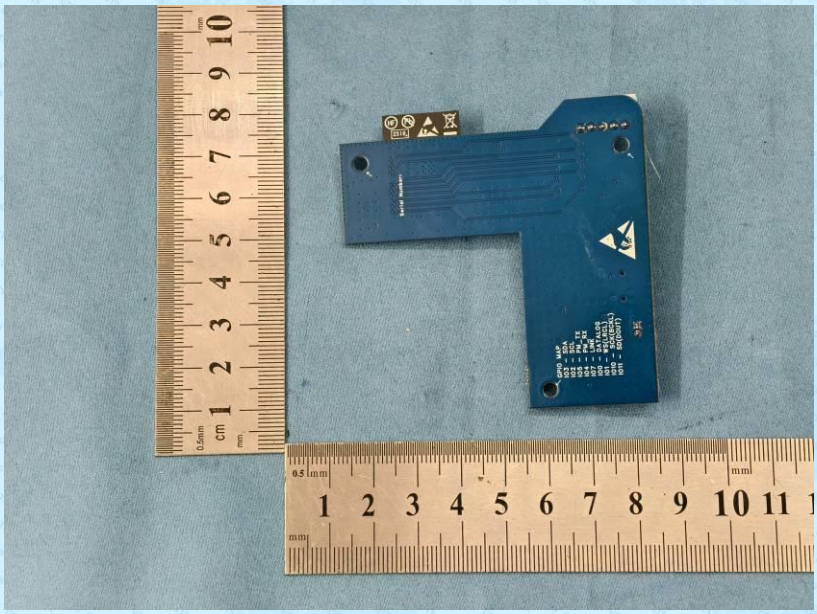












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